

(ii) Principle of Duality $\rightarrow$ If $<$ is a partial order relation on any set, then its inverse relation $>$ is also a partial order relation, known as the Dual order

* We can obtain the Dual of a lattice by interchanging $<$ with $>$ and $\wedge$ with $\vee$

Q Let $A = \{1, 2, 3, 4, 5\}$ and $(A, \leq)$ be a lattice. Then the lattice $(A, \geq)$ is the dual of $(A, \leq)$

(iii) Lattice as an Algebraic System $\rightarrow$

A lattice $(L, \vee, \wedge)$ is an algebraic system because L together with two binary operations $\vee$ and $\wedge$ satisfies certain algebraic properties.

Theorem $\rightarrow$ Let $(L, \vee, \wedge)$ be a lattice. Then the following properties are satisfied for all $x, y, z \in L$

(a) Commutative law

$$(i) \quad (x \vee y) = (y \vee x)$$

$$(ii) \quad (x \wedge y) = (y \wedge x)$$

(b) Associative law

(i)

$$x \vee (y \vee z) = (x \vee y) \vee z$$

(ii)

$$x \wedge (y \wedge z) = (x \wedge y) \wedge z$$

(c)

Absorption law

(i)

$$x \vee (x \wedge y) = x$$

(ii)

$$x \wedge (x \vee y) = x$$

(d) Idempotent law

$$(i) \quad x \vee x = x$$

$$(ii) \quad x \wedge x = x$$

(iv) Types of lattices

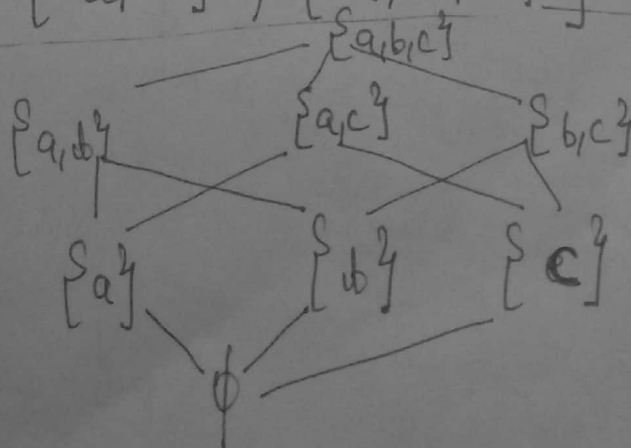
1 Distributive lattice $\rightarrow$ A lattice $(L, <)$ is called a distributive lattice iff the distributive laws hold, that is $\forall x, y, z \in L$ we have

$$(a) \quad x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$$

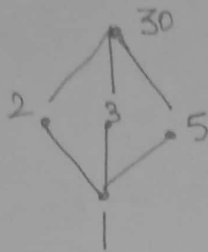
$$(b) \quad x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$$

Eg $(P(S), \subseteq)$ where $S = \{a, b, c\}$ is a distributive lattice.

$$P(S) = \{ \phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\} \}$$



Ex. Show that the lattice $L = \{1, 2, 3, 5, 30\}$ is non-distributive under the partial order relation $|$



The least element is 1 and greatest element is 30.

$$2 \wedge (3 \vee 5) = 2 \wedge 30 = 2$$

$$(2 \wedge 3) \vee (2 \wedge 5) = 1 \vee 1 = 1$$

$$2 \wedge (3 \vee 5) \neq (2 \wedge 3) \vee (2 \wedge 5)$$

$\therefore$ Not Distributive

* Every chain is a distributive lattice.

⑧ Bounded Lattice $\rightarrow$ A lattice L is said to be Bounded if it has a greatest element as well as a least element.

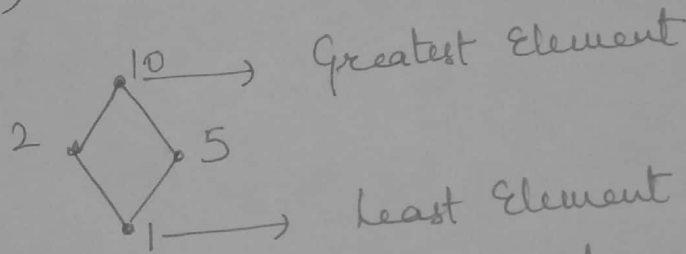
* The greatest element is denoted by 1, which is often called the unit element and the least element is denoted by 0, which is often called the zero element

* Hence, if $(L, \leq)$ is a Bounded lattice then,
 $\forall x \in L, 0 < x < L$ and

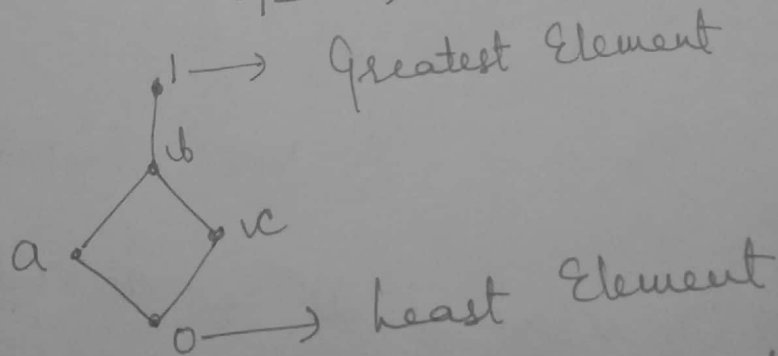
$$x \vee 0 = x, \quad x \vee 1 = 1$$

$$x \wedge 0 = 0, \quad x \wedge 1 = x$$

Eg 1 Lattice $(L, \leq)$ where $L = \{1, 2, 5, 10\}$



2.



* The lattice formed by the set $\mathbb{Z}$ of integers under the partial order relation $\leq$ is not bounded because it has neither a greatest nor a least element.

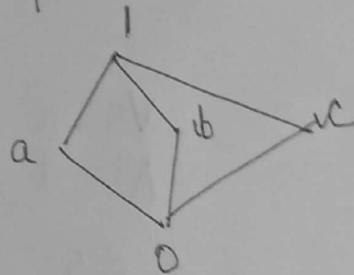
Complemented Lattice

Let L be a bounded lattice with the greatest element 1 and the least element 0 . Let $x \in L$.
Then $x' \in L$, is called a complement of x if,
 $x \vee x' = 1$ and $x \wedge x' = 0$.

→ 1 and 0 are complements of each other

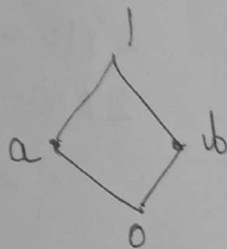
→ An element can have more than 1 complements.

eg. (1)



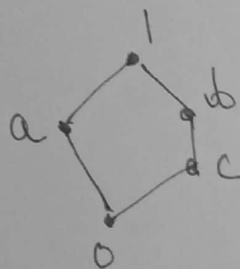
$$\begin{aligned} a' &= \{b, c\} & b' &= \{a, c\} \\ a \wedge b &= 0 & a \vee b &= 1 \\ a \wedge c &= 0 & a \vee c &= 1 \end{aligned}$$

(2)



$$\begin{aligned} a' &= b \\ b' &= a \end{aligned}$$

(3)



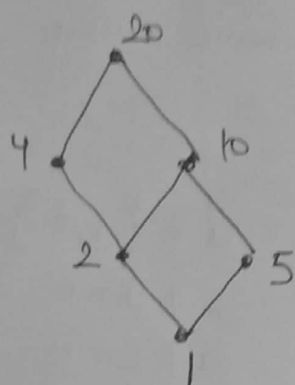
$$\begin{aligned} a' &= \{b, c\} \\ b' &= \{a\} \\ c' &= \{a\} \end{aligned}$$

→ Complement of an element is never in the same branch

Ex 4

Find the complements of each elements of the lattice D_{20} , where D_{20} is the set of divisors of 20.

$$\rightarrow D_{20} = \{1, 2, 4, 5, 10, 20\}$$



$$1' = 20$$

$$20' = 1$$

$$2' = ?$$

$$2 \vee 5 = 10 \neq 20$$

$$2 \wedge 1 = 1$$

$$\left[\begin{array}{l} \text{LCM}(2, 5) = 10 \\ \text{GCD}(2, 5) = 1 \end{array} \right]$$

$$\therefore 2' \neq 5$$

$$2 \vee 10 = 20$$

$$2 \wedge 10 = 2 \neq 1$$

$$\therefore 2' \neq 10$$

$$\left[\begin{array}{l} \text{LCM}(2, 10) = 20 \\ \text{GCD}(2, 10) = 2 \end{array} \right]$$

$$4' = ?$$

$$4 \vee 5 = 20$$

$$4 \wedge 5 = 1$$

$$\therefore 4' = 5$$

$$5' = 4$$

2 and 10 do not have a complement

$\therefore$ Given lattice is Not Complemented lattice.

Alternate Technique

V	1	2	4	5	10	20
1	1	2	4	5	10	20^*
2	2	2	4	10	10	20^*
4	4	4	4	20^*	20^*	20^*
5	5	10	20^*	5	10	20^*
10	10	10	20^*	10	10	20^*
20	20^*	20^*	20^*	20^*	20^*	20^*

A	1	2	4	5	10	20
1	1+	1+	1+	1+	1+	1^+
2	1+	2	2	1+	2	2
4	1+	2	4	1^+	2	4
5	1+	1+	1^+	5	5	5
10	1+	2	2	5	10	10
20	1^+	2	4	5	10	20

→ 1 has a * and + match corresponding to 20
 If 20 has a * and + match corresponding to 1
 $\therefore 1' = 20$ and $20' = 1$

→ If 4 has a * and + match corresponding to 5
 5 has a * and + match corresponding to 4
 $\therefore 4' = 5$ and $5' = 4$

→ No element for 2 and 10 have a * and + mark at the same position.